Module 15 - chapter 16

Project: Run Docker applications - Part 1

The project files used in this lecture can be found here: https://gitlab.com/twn-devops-bootcamp/latest/15-ansible/terraform-learn/-/tree/feature/deploy-to-ec2

Ansible & Docker

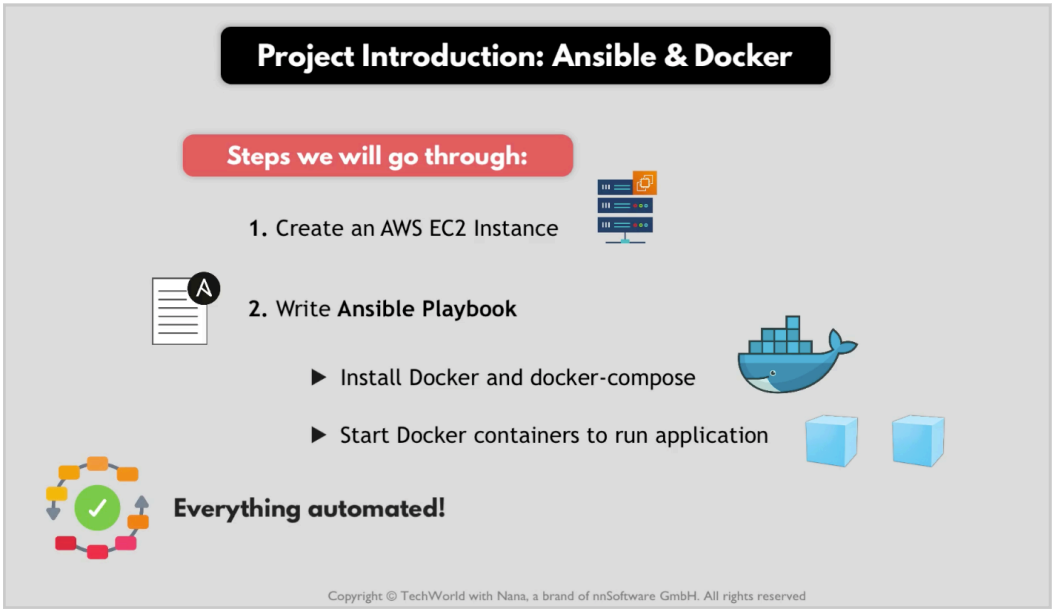
Project Overnieuw

We will write an ansible playbook to do the below:

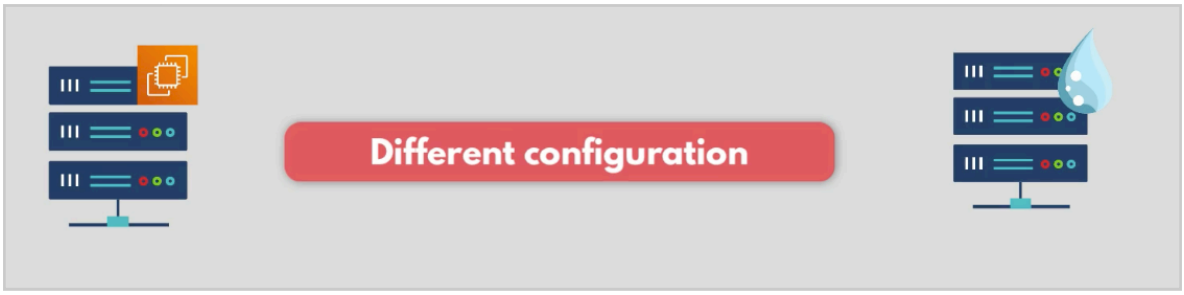
- Install Docker
- Install Docker compose

On a AWS EC2 server.

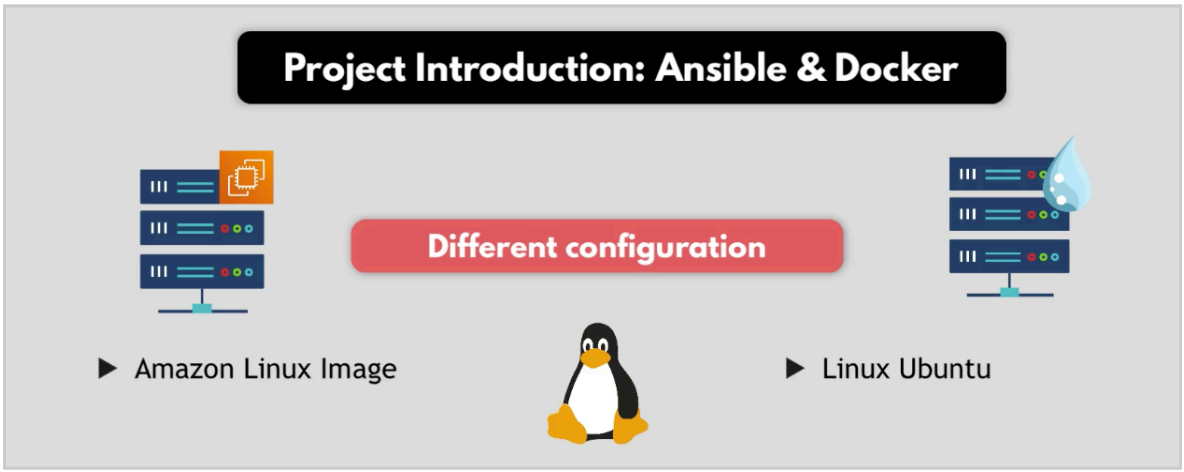
- Using a docker compose file from one of our previous projects we will start the containers on that EC2 server.



Configuration for the AWS EC2 server is different to the droplet configuration.



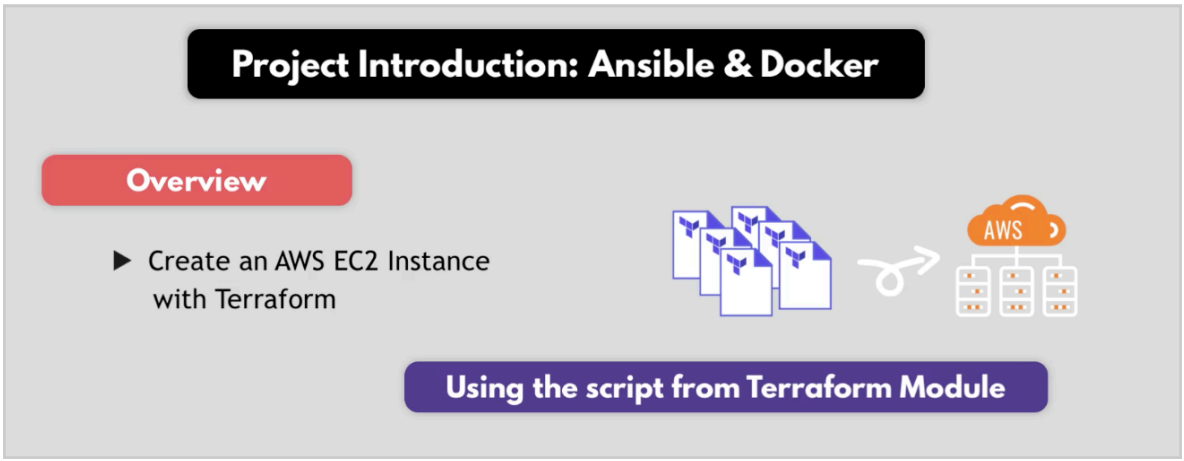
Their OS are different.



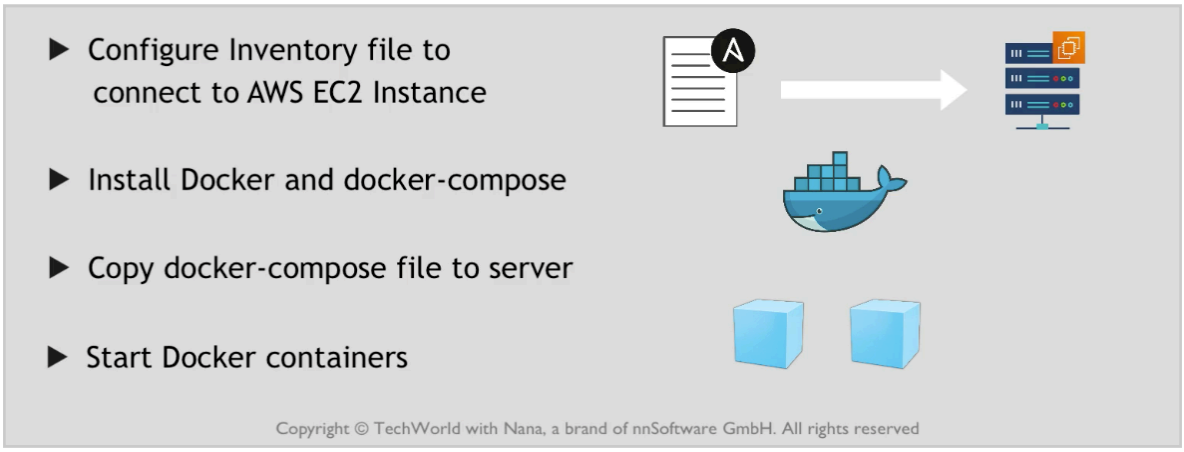
EC2 server uses Amazon Linux, the droplet uses Linux Ubuntu.

So installing packages and other things will be different.

1. To create the EC2 instance we will use Terraform so we do it automatically. We will use a tf file from a previous project



2. Connect EC2 instance to Ansible and create a playbook to install docker and docker compose, copy docker-compose file to server and start docker containers



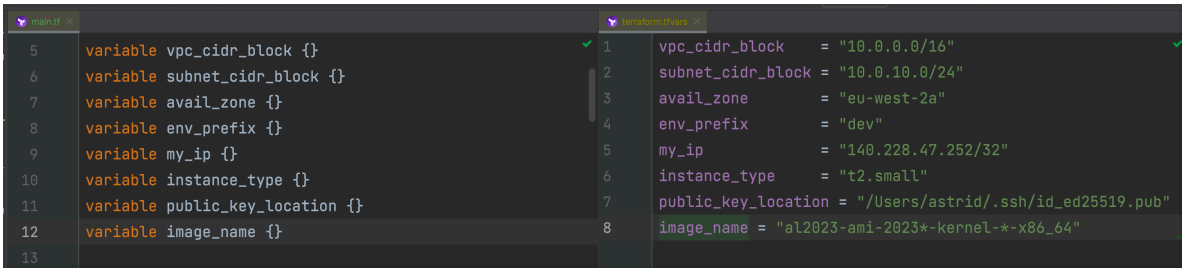
Create AWS EC2 Instance

I copy the code from Module 12 - chapter 14 https://github.com/AstridCaballero/Module_12_Terraform/blob/Module_12/Terraform_chapter-14/main.tf

And create a new branch https://github.com/AstridCaballero/Module_15_Ansible/tree/Module_15/Ansible_chapter-16_Terraform

And my branch doesn't have terraform.tfvars committed to the remote branch, so I found them in my notes and here they are:

```
vpc_cidr_block    = "10.0.0.0/16"
subnet_cidr_block = "10.0.10.0/24"
avail_zone        = "eu-west-2a"
env_prefix        = "dev"
my_ip             = "140.228.47.252/32"
instance_type     = "t2.small"
public_key_location = "/Users/astrid/.ssh/id_ed25519.pub"
image_name = "al2023-ami-2023*-kernel-*-x86_64"
```



In the previous terraform projects we used a shell script

main.tf | entry-script.sh

```
#!/bin/bash
sudo dnf update -y && sudo dnf install -y docker
sudo systemctl start docker
sudo usermod -aG docker ec2-user
docker run -p 8080:80 nginx
```

Here we don't need it, because we only want to provision the server, we don't want to execute anything inside them yet. We want to use Ansible to configure the EC2 server.

So we remove the shell file and we also remove lines 139 and 141 👇 when creating the resource "aws_instance"

```
127
128  resource "aws_instance" "myapp-server" {
129    ami = data.aws_ssm_parameter.latest-amazon-linux-image.value
130    instance_type = var.instance_type
131
132    subnet_id = aws_subnet.myapp-subnet-1.id
133    vpc_security_group_ids = [aws_default_security_group.default-sg.id]
134    availability_zone = var.avail_zone
135
136    associate_public_ip_address = true
137    key_name = aws_key_pair.ssh-key.key_name
138
139    user_data = file("entry-script.sh")
140
141    user_data_replace_on_change = true
142
143    tags = {
144      Name = "${var.env_prefix}-server"
145    }
146  }
```

Now we can execute the configuration
-> terraform init

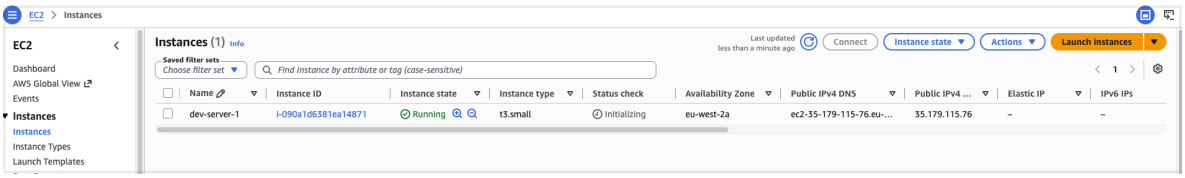
```
❯ terraform init
Initializing provider plugins found in the configuration...
- Finding hashicorp/aws versions matching "~> 6.0"...
- Installing hashicorp/aws v6.58.0...
- Installed hashicorp/aws v6.58.0 (signed by HashiCorp)

Initializing the backend...


Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

-> terraform apply

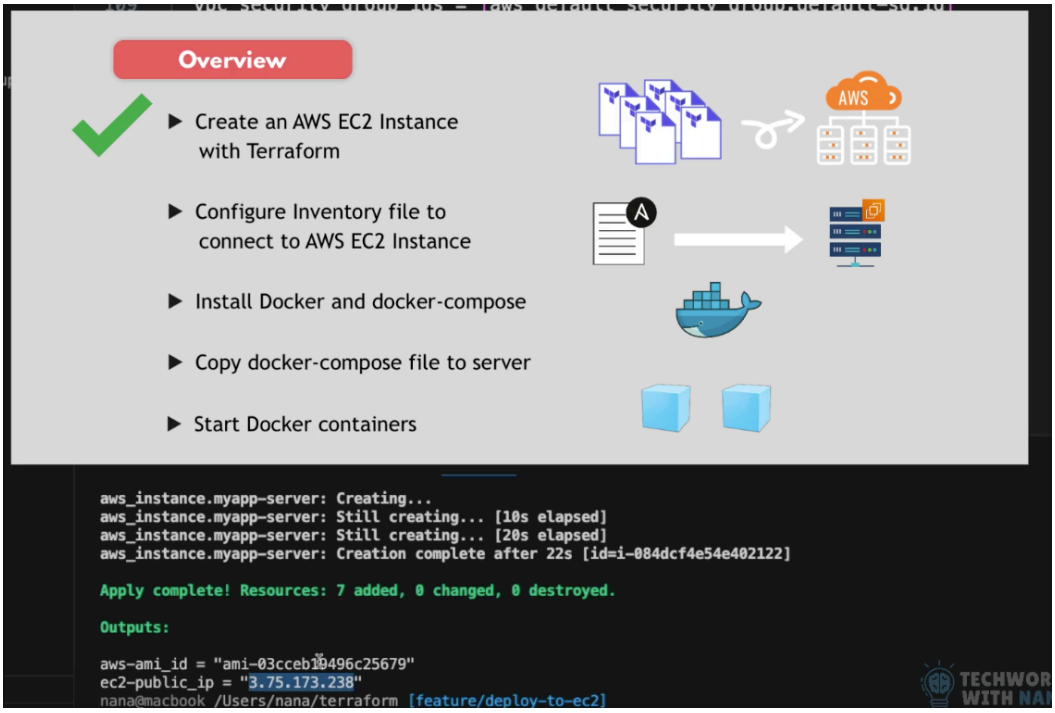
(I had to run it twice, first run created everything but the EC2 instance, second run I edited t2.small to t3.small)

```
aws_instance.myapp-server-one: Creating...
aws_instance.myapp-server-one: Still creating... [00m10s elapsed]
aws_instance.myapp-server-one: Creation complete after 13s [id=i-090a1d6381ea14871]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

aws-ami_id = "ami-01c952cfc86b7870d"
ec2-public_ip_1 = "35.179.115.76"
```

And we have the EC2 instance running



Now we have a server that it has now configuration yet



Now Ansible will take from here ✅

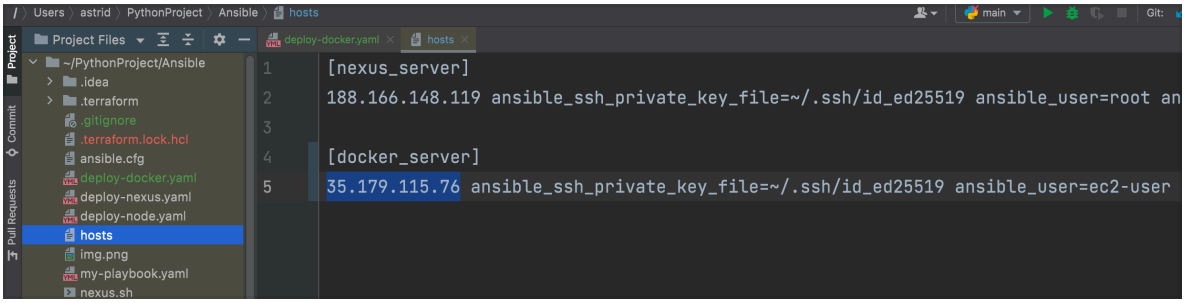
Adjust Inventory File

I create a new branch from Module 15 - chapter 15 https://github.com/AstridCaballero/Module_15_Ansible/tree/Module_15/Ansible_chapter-16_ansible

From the terraform execution:

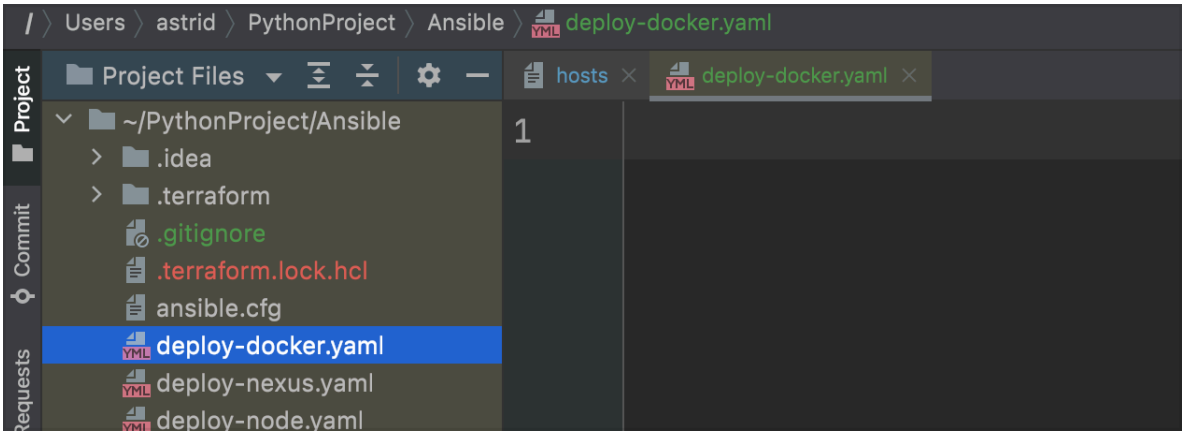
```
aws_instance.myapp-server-one: Creating...
aws_instance.myapp-server-one: Still creating... [00m10s elapsed]
aws_instance.myapp-server-one: Creation complete after 13s [id=i-090a1d6381ea14871]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

aws-ami_id = "ami-01c952cfc86b7870d"
ec2-public_ip_1 = "35.179.115.76"
```

I copy and I add the IP address of the EC2 instance to the ‘hosts’ file



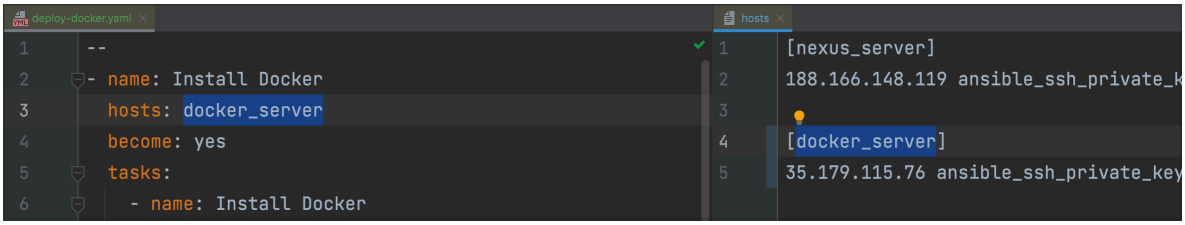
And for Ec2 instances the user is ec2-user so I use that value for ansible_user:

And let's create a new playbook 'deploy-docker.yaml'



Write first play

Install docker



Now in the EC2 instance the package manager tool is not 'apt', so if I ssh into the ec2 instance and check for 'apt' it can't be found

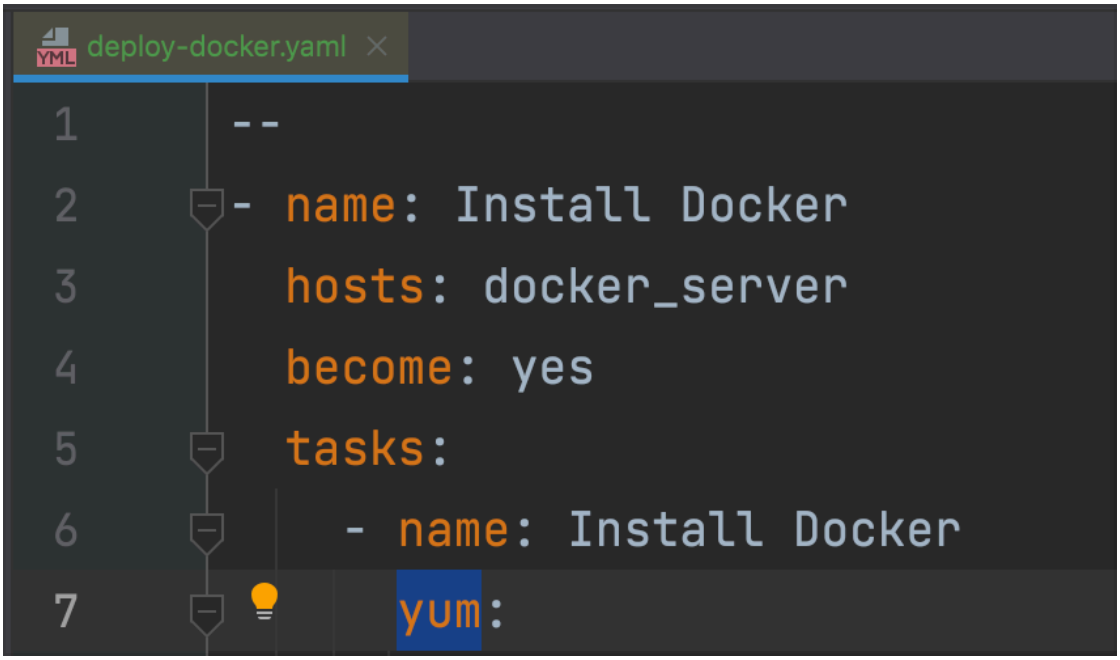
```
❯ ssh ec2-user@35.179.115.76
The authenticity of host '35.179.115.76 (35.179.115.76)' can't be established.
ED25519 key fingerprint is SHA256:br4jN6Y2eL2gwss4BmFO8O0QdYEFMTZf3NEQTp3bJz4.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '35.179.115.76' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-10-56 ~]$ apt
-bash: apt: command not found
[ec2-user@ip-10-0-10-56 ~]$
```

And if I try ‘yum’ we can see is installed

```
[ec2-user@ip-10-0-10-56 ~]$ apt
-bash: apt: command not found
[ec2-user@ip-10-0-10-56 ~]$ yum
usage: yum [options] COMMAND

List of Main Commands:

alias                    List or create command aliases
autoremove               remove all unneeded packages that were originally inst
check                    check for problems in the packagedb
check-update             check for available package upgrades
clean                    remove cached data
deplist                  [deprecated, use repoquery --deplist] List package's d
distro-sync              synchronize installed packages to the latest available
downgrade                Downgrade a package
group                    display, or use, the groups information
help                     display a helpful usage message
history                  display, or use, the transaction history
```

So yum is the package manager tool in EC2 instance!

And now our play looks like this:

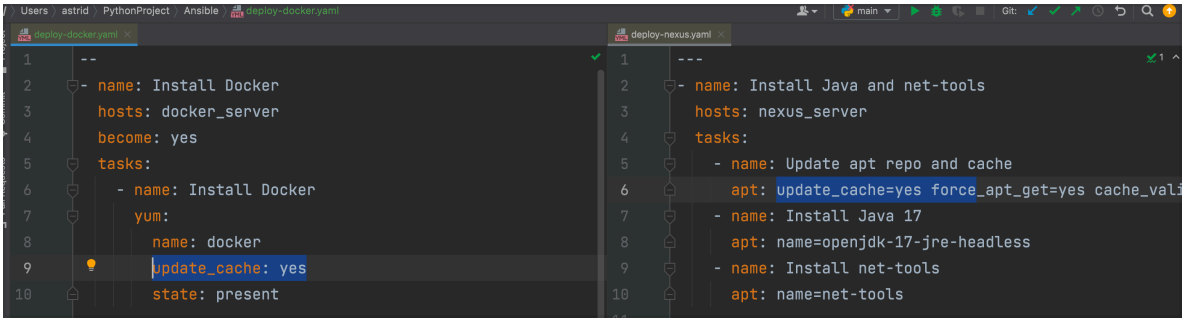


And in ansible ‘yum’ module docs
https://docs.ansible.com/projects/ansible/devel/collections/ansible/builtin/yum_repository_module.html
We can find example of how to use it.

Examples

```yaml
- name: Add repository
  ansible.builtin.yum_repository:
    name: epel
    description: EPEL YUM repo
    baseurl: https://download.fedoraproject.org/pub/epel/$releasever/$basearch/

- name: Add multiple repositories into the same file (1/2)
  ansible.builtin.yum_repository:
    name: epel
    description: EPEL YUM repo
    file: external_repos
    baseurl: https://download.fedoraproject.org/pub/epel/$releasever/$basearch/
    gpgcheck: no

- name: Add multiple repositories into the same file (2/2)
  ansible.builtin.yum_repository:
    name: rpmforge
    description: RPMforge YUM repo
    file: external_repos
    baseurl: http://apt.sw.be/redhat/el7/en/$basearch/rpmforge
    mirrorlist: http://mirrorlist.repoforge.org/el7/mirrors-rpmforge
    enabled: no

# Handler showing how to clean yum metadata cache
- name: yum-clean-metadata
  ansible.builtin.command: yum clean metadata

# Example removing a repository and cleaning up metadata cache
- name: Remove repository (and clean up left-over metadata)
  ansible.builtin.yum_repository:
    name: epel
    state: absent
  notify: yum-clean-metadata

- name: Remove repository from a specific repo file
  ansible.builtin.yum_repository:
    name: epel
    file: external_repos
    state: absent
```

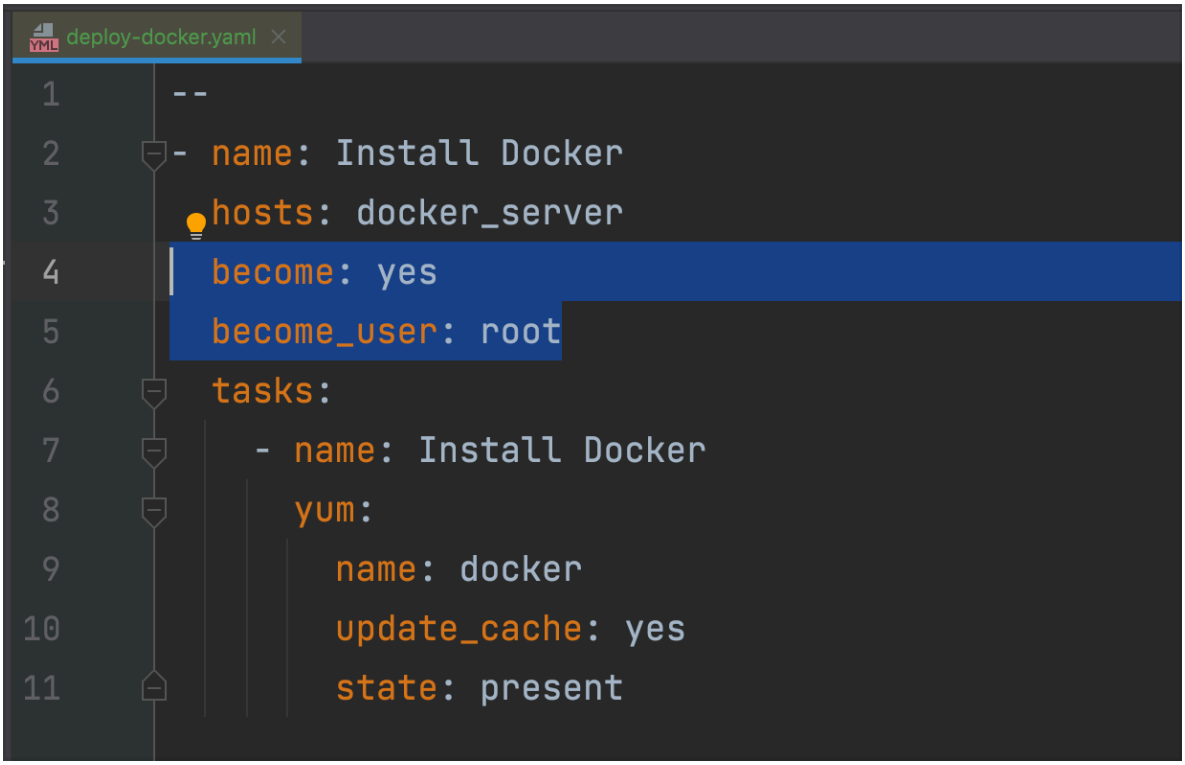
And we want to install docker using ‘yum’ and do something similar to what we did for ‘deploy-nexus.yaml’



present *to install a package, absent to uninstall* 👆

Currently we are using ‘ec2-user’ which is an ‘admin’ user not ‘root’ user. And to install a package we need to use a ‘root’ user:

So we need to tell ansible to run the install command using ‘root’ user instead of ‘ec2-user’ and for that we use ‘become’



And the default value of ‘become_user’ is root so we actually don’t need line 5 👆 as it is redundant

deploy-docker.yaml

```yaml
---
- name: Install Docker
  hosts: docker_server
  become: yes
  tasks:
    - name: Install Docker
      yum:
        name: docker
        update_cache: yes
        state: present
```

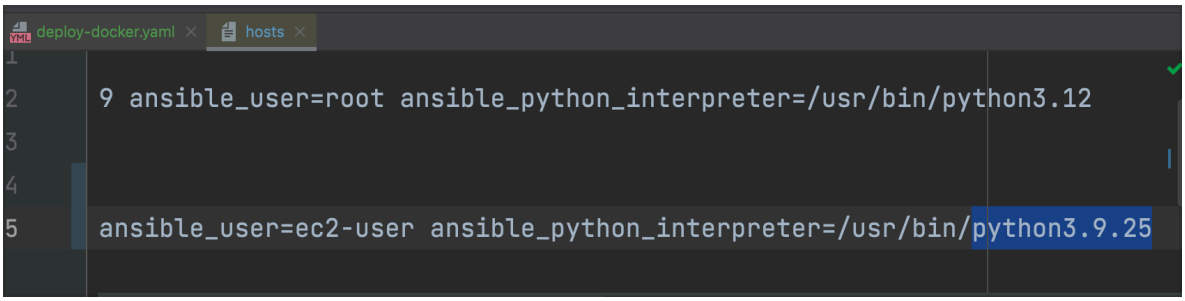
Let's execute the playbook
-> ansible-playbook deploy-docker.yaml

```
TASK [Gathering Facts] *********************************************************
[ERROR]: Task failed: Action failed: The following modules failed to execute: ansible.le

Task failed: Action failed.

<<< caused by >>>

The following modules failed to execute: ansible.legacy.setup.

+--[ Sub-Event 1 of 1 ]---
|
| The module interpreter '/usr/bin/python3.12' was not found. Consider overriding the co
. See stdout/stderr for the returned output.
|
+--[ End Sub-Event ]---
```

I got an error related to python's version, so I checked the Ec2 python 3 version and is not 3.12, it is 3.9.25

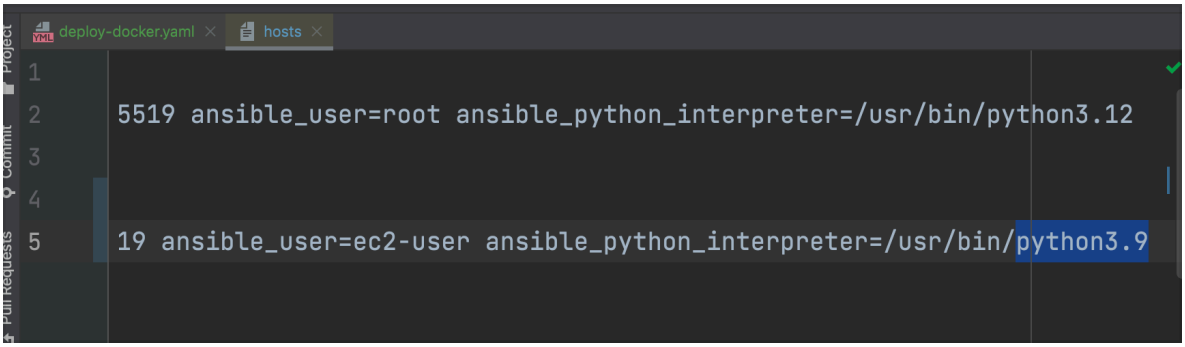
```
[ec2-user@ip-10-0-10-56 ~]$ python3 --version
Python 3.9.25
[ec2-user@ip-10-0-10-56 ~]$
```

So I update the value in ‘hosts’ value



Same error. The problem is that I need the binary filename not the full version string (major.minor.patch).
 Binaries are only ever named after major.minor — python3.9, not python3.9.25. The patch version (25) isn't part of the executable's name on disk.

So I edit the ‘hosts’ file again to use 3.9 only



And it works

```
❯ ansible-playbook deploy-docker.yaml

PLAY [Install Docker] **********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [35.179.115.76]

TASK [Install Docker] **********************************************************************************************************
changed: [35.179.115.76]

PLAY RECAP *********************************************************************************************************************
35.179.115.76              : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Write 2nd play

Install docker-compose

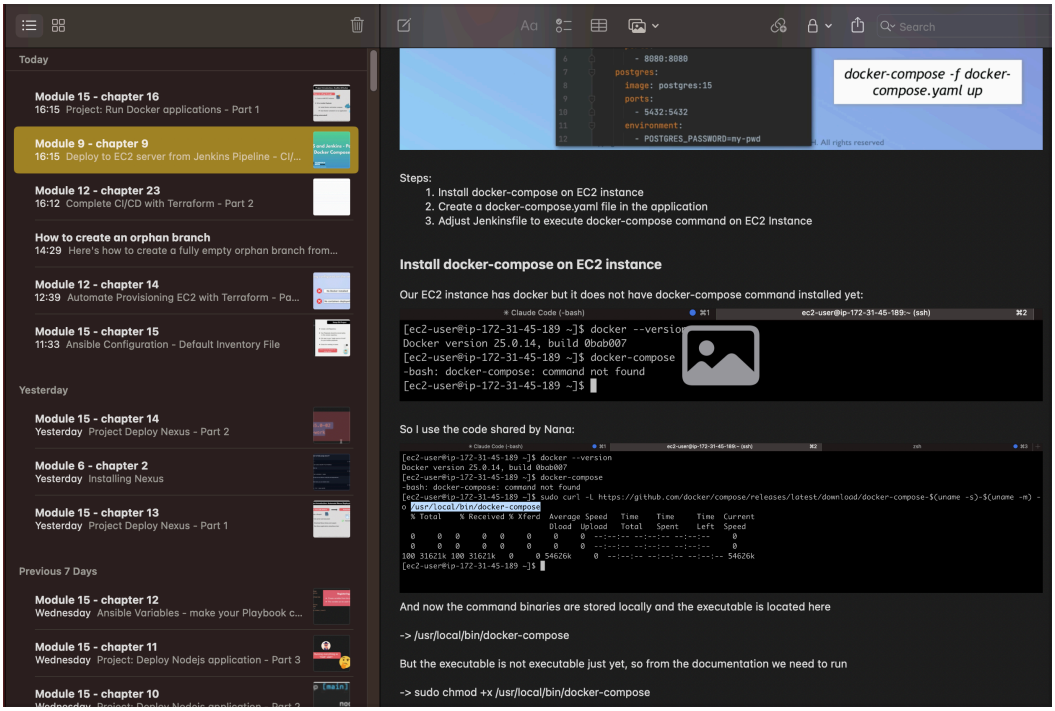
So we want to start containers from the docker-compose file on the EC2 server and for that we need both docker and docker -compose

In EC2 we can't install docker compose using yum because docker compose is not a yum package ‼️

If we try to do it manually it fails:

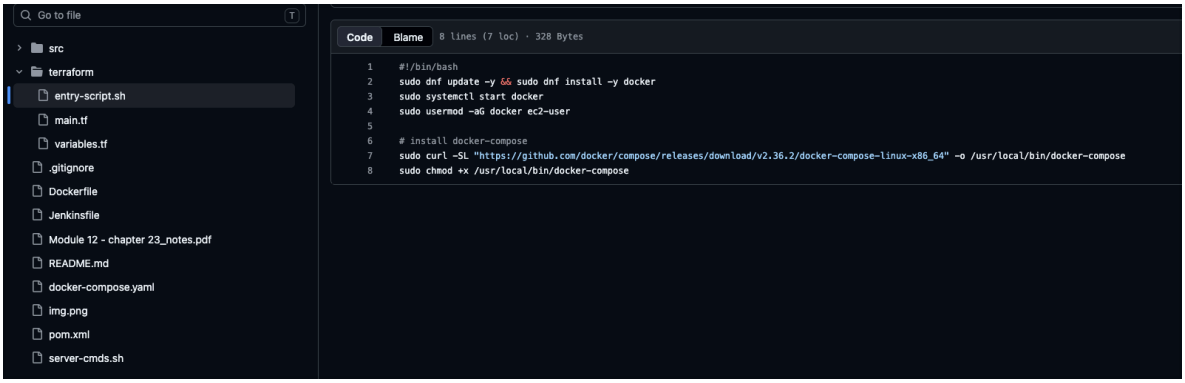
```
[ec2-user@ip-10-0-10-186 ~]$ sudo yum install docker-compose
Last metadata expiration check: 1:00:15 ago on Thu Feb 15 09:21:52 2024.
No match for argument: docker-compose
Error: Unable to find a match: docker-compose
[ec2-user@ip-10-0-10-186 ~]$
```

In a previous demo we installed docker-compose directly into a EC2 server using a CI/CD pipeline in Module 12 - chapter 23
With similar steps to what we did in Module 9 - chapter 9
https://github.com/AstridCaballero/Module_9_Java-maven-app/blob/Module_9/chapter_9_Jenkins-Jobs_Docker_Compose/server-cmds.sh

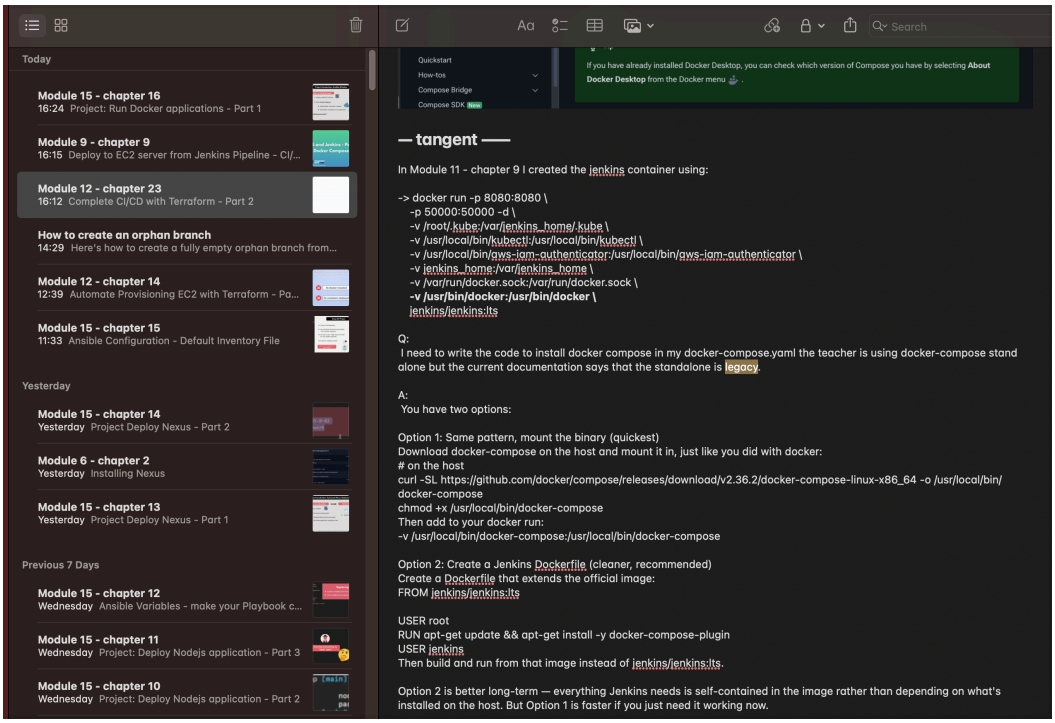


Then in Module 12 - chapter 23 we used an entry-script.sh that was executed in a fresh new EC2 instance that we provisioned with terraform.

https://github.com/AstridCaballero/Module_12_Terraform/blob/Module_12/Terraform_chapter-23/terraform/entry-script.sh

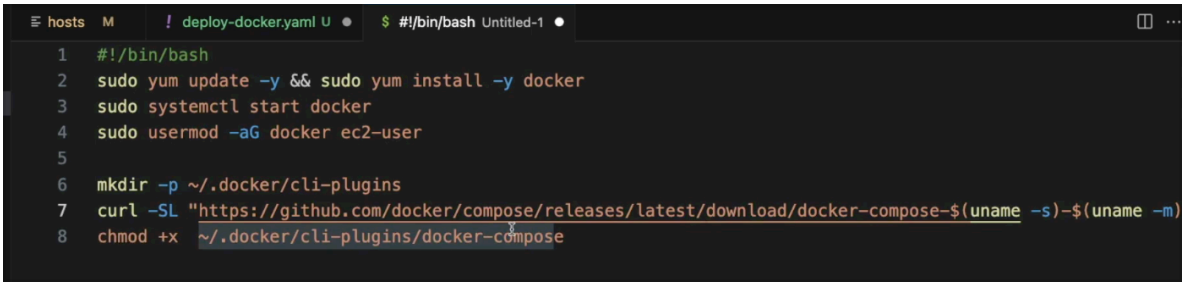


And Nana installed docker-compose as a stand alone installation but in Module 12 - chapter 23 I couldn't because at the time I did the demo is was already legacy as seen in the notes:



Now for this demo, Nana won't use the stand alone installation but the cli plugin installation for docker-compose which works more intuitively with Ansible's docker modules.

So these are the commands with the addition of line 6 and the editions of some paths from Nana for this demo that we need to translate to ansible:

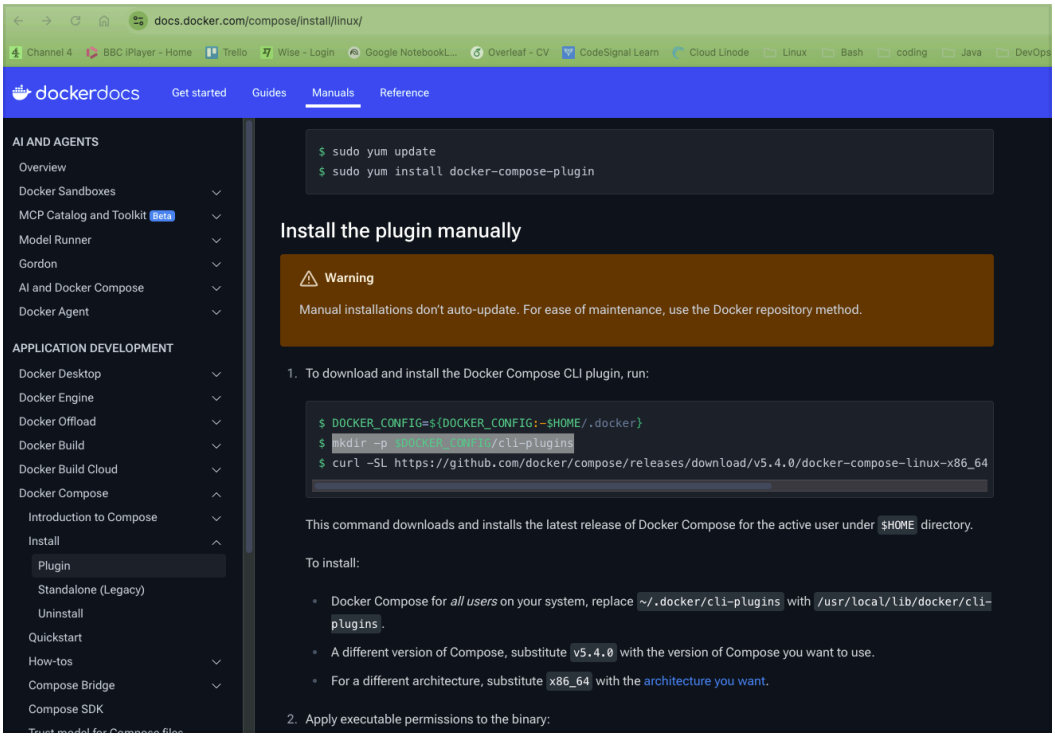


Line 7 👆 is long, here is the rest:

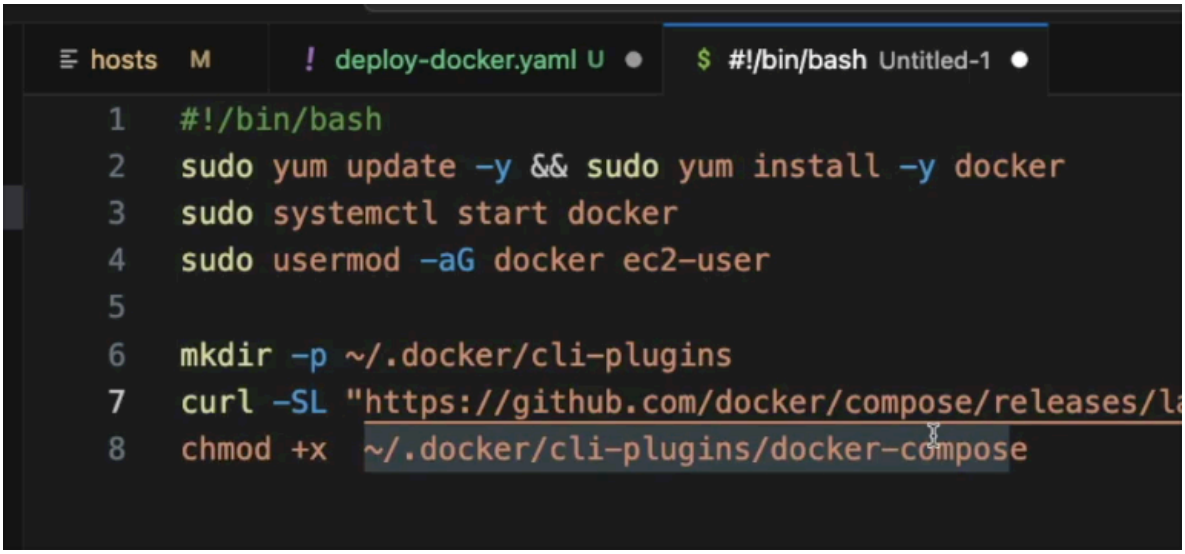
And Nana is following the plugin installation manually, and for that we need to create a directory Called ‘cli-plugins’ first which is in line 6 of her shell file above.

https://docs.docker.com/compose/install/linux/

We need this folder to download docker-compose in that location as per line 7 in the shell file



And finally we need to make the docker-compose file executable as per line 8 in the shell file.



Now let's map the shell commands into an ansible play:

```
12    - name: Install Docker-compose
13      hosts: docker_server
14      tasks:
15        - name: Create docker-compose directory
16          file:
```

And we will use the ‘file’ module to create the new directory

From the documentation we know that the new folder needs to be created inside the /.docker folder

```
$ DOCKER_CONFIG=${DOCKER_CONFIG:-$HOME/.docker}
$ mkdir -p $DOCKER_CONFIG/cli-plugins
```

And the folder name will be ‘cli-plugins’

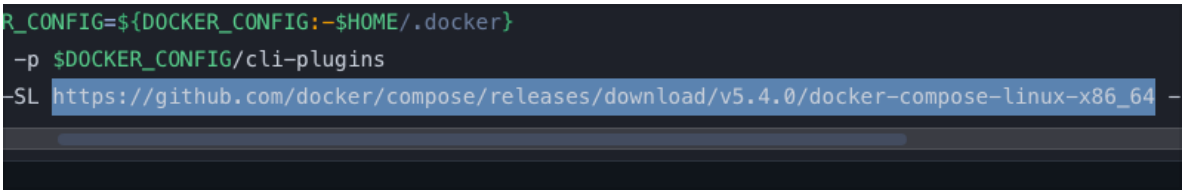
```
12    - name: Install Docker-compose
13      hosts: docker_server
14      tasks:
15        - name: Create docker-compose directory
16          file:
17            path: ~/.docker/cli-plugins
18            state: directory
```

Now let’s download from internet and install docker-compose so we will use ‘get_url’ module

Nana uses this url:

```
19  - name: Install docker-compose
20    get_url:
21      url: https://github.com/docker/compose/releases/latest/download/docker-compose-$(uname -s)-$(uname -m)
22
```

So I compared with the documentation and it looks a bit different:



And AI says:
here's what's actually going on:

Screenshot from Nana's code: grabs the latest release and tries to build the filename dynamically:
.../releases/latest/download/docker-compose-$(uname -s)-$(uname -m) -o ~/.docker/cli-plugins/docker-compose

Screenshot from docs: pins an exact version and hardcodes the OS/arch:
.../compose/releases/download/v5.4.0/docker-compose-linux-x86_64 -o ...

The dynamic version is the "official-looking" pattern people copy around, but there's a real gotcha: Docker Compose v2's release assets use lowercase OS names (docker-compose-linux-x86_64), while uname -s on Linux returns Linux with a capital L. So $(uname -s) produces Linux, not linux, and the built URL won't match any actual file on GitHub — it'll 404.

Worse: if your curl command doesn't include -f (fail on HTTP error), curl will happily save the GitHub 404 error page as if it were the binary. It'll "succeed" silently, and then docker compose version will fail or error with something cryptic later — much harder to debug than an obvious curl failure.

Two ways to fix it if you want to keep the dynamic version:

curl -SL https://github.com/docker/compose/releases/latest/download/docker-compose-$(uname -s | tr '[:upper:]' '[:lower:]')-$(uname -m) -o ~/.docker/cli-plugins/docker-compose

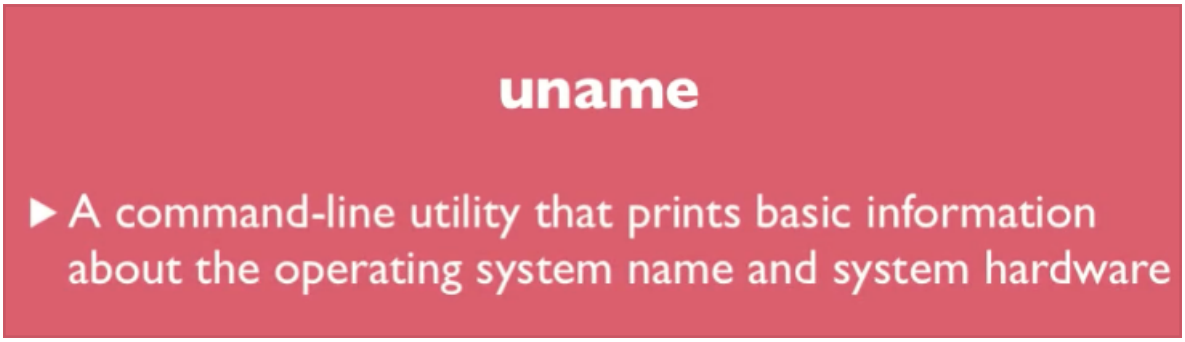
Or just be explicit like Screenshot from docs does — hardcode linux-x86_64 (fine since you know your EC2 target is x86_64 Linux) and pin a version for reproducibility. That's honestly the safer choice for a provisioning script: predictable, reproducible, no silent case-mismatch failures.

So I use the hardcoded url provided by the documentation

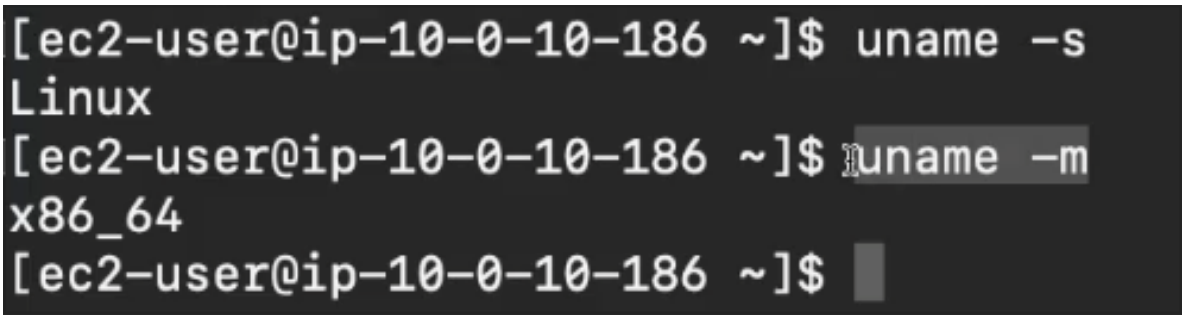
```yaml
- name: Install Docker-compose
  hosts: docker_server
  tasks:
    - name: Create docker-compose directory
      file:
        path: ~/.docker/cli-plugins
        state: directory
    - name: Install docker-compose
      get_url:
        url: https://github.com/docker/compose/releases/download/v5.4.0/docker-compose-linux-x86_64
        dest: ~/.docker/cli-plugins/docker-compose
        mode: +x
```

And the mode we set it to executable.

uname is a command:



We execute them in the terminal of the server and we can get:



We know uname -m is x86_64 which is the architecture of the server as the AI mentioned before, but if we don't then we can actually create a task before installing docker-compose task and store it in a variable and the use that variable

```yaml
- name: Install Docker-compose
  hosts: docker_server
  tasks:
    - name: Create docker-compose directory
      file:
        path: ~/.docker/cli-plugins
        state: directory
    - name: get architecture of remote machine
      shell: uname -m
      register: remote_arch
    - name: Install docker-compose
      get_url:
        url: https://github.com/docker/compose/releases/download/v5.4.0/docker-compose-linux-{{remote_arch.stdout}}
        dest: ~/.docker/cli-plugins/docker-compose
        mode: +x
```

Pattern	Behavior
/releases/download/v5.4.0/docker-compose-linux-x86_64	Pinned — always fetches that exact tagged release
/releases/latest/download/docker-compose-linux-x86_64	Dynamic — redirects to whatever the current latest release is

So if you want "always latest," drop the version segment entirely:

https://github.com/docker/compose/releases/latest/download/docker-compose-linux-{{ remote_arch.stdout }}

If you want reproducible/pinned installs (recommended for provisioning scripts — so a fresh EC2 build next month doesn't silently pick up a newer, possibly breaking, Compose version), keep v5.4.0 hardcoded as you have it.

So I decided to create a variable to hold the version of docker-compose

deploy-docker.yaml

```yaml
name: Install Docker-compose
hosts: docker_server
vars_files:
  project-vars
tasks:
  - name: Create docker-compose directory
    file:
      path: ~/.docker/cli-plugins
      state: directory
  - name: get architecture of remote machine
    shell: uname -m
    register: remote_arch
  - name: Install docker-compose
    get_url:
      url: https://github.com/docker/compose/releases/download/v{{docker_compose_version}}/docker-compo
```

Document 1/1 › Item 2/2 › tasks: › Item 3/3 › get_url: › url: › https://github.com/d...

project-vars

```yaml
version: 1.0.0
location: /Users/astrid/PythonProject/Ansible/nodejs-app
linux_name: astrid
user_home_dir: /home/{{linux_name}}

nexus_version: 3.84.1-01
docker_compose_version: 5.4.0
```

See I also had to add lines 14 and 15 👆

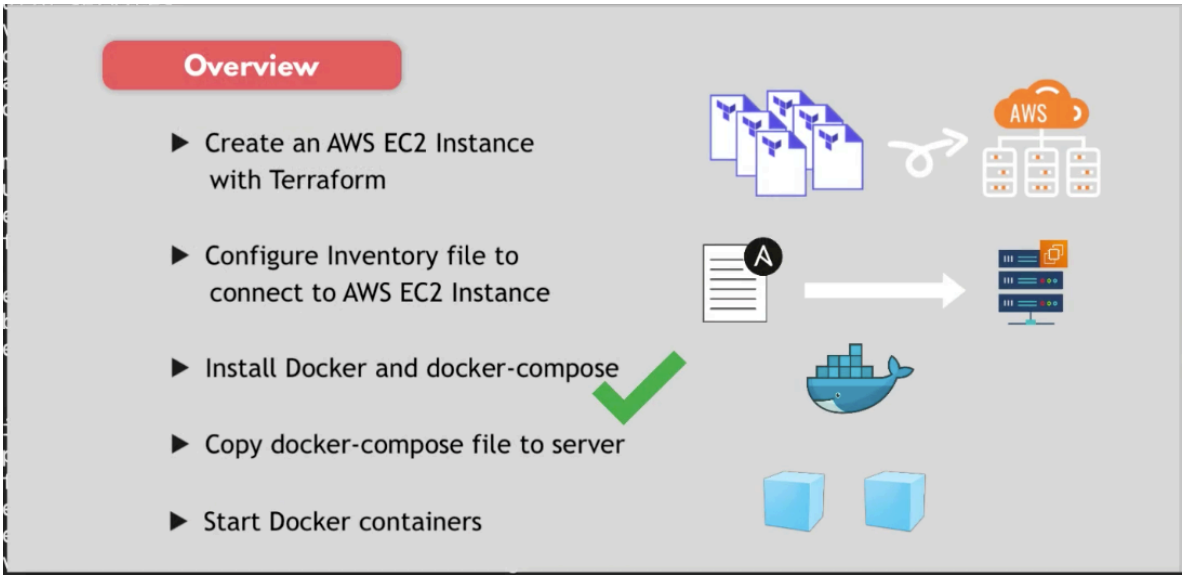
Run the playbook
-> ansible-playbook deploy-docker.yaml

```
TASK [Gathering Facts] *********************************************************************************************
ok: [35.179.115.76]

TASK [Create docker-compose directory] *****************************************************************************
changed: [35.179.115.76]

TASK [get architecture of remote machine] **************************************************************************
changed: [35.179.115.76]

TASK [Install docker-compose] **************************************************************************************
changed: [35.179.115.76]

PLAY RECAP *********************************************************************************************************
35.179.115.76              : ok=6    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

And if I check the server

```
[ec2-user@ip-10-0-10-56 ~]$ docker compose
Usage:  docker compose [OPTIONS] COMMAND

Define and run multi-container applications with Docker

Options:
      --all-resources              Include all resources, even those not
      --ansi string                Control when to print ANSI control cha
      --compatibility              Run compose in backward compatibility
      --dry-run                    Execute command in dry run mode
      --env-file stringArray       Specify an alternate environment file
  -f, --file stringArray           Compose configuration files
      --parallel int               Control max parallelism, -1 for unlimi
      --profile stringArray        Specify a profile to enable
      --progress string            Set type of progress output (auto, tty
      --project-directory string   Specify an alternate working directory
                                   (default: the path of the, first speci
  -p, --project-name string        Project name

Management Commands:
  bridge                  Convert compose files into another model

Commands:
  attach                  Attach local standard input, output, and error
  build                   Build or rebuild services
  commit                  Create a new image from a service container's c
```

Now we have the command available



From this commands:

Code Blame 8 lines (7 loc) · 328 Bytes

```
1  #!/bin/bash
2  sudo dnf update -y && sudo dnf install -y docker
3  sudo systemctl start docker
4  sudo usermod -aG docker ec2-user
5
6  # install docker-compose
7  sudo curl -SL "https://github.com/docker/compose/releases/download/v2.36.2/docker-compose-linux-x86_64" -o /usr/local/bin/docker-compose
8  sudo chmod +x /usr/local/bin/docker-compose
```

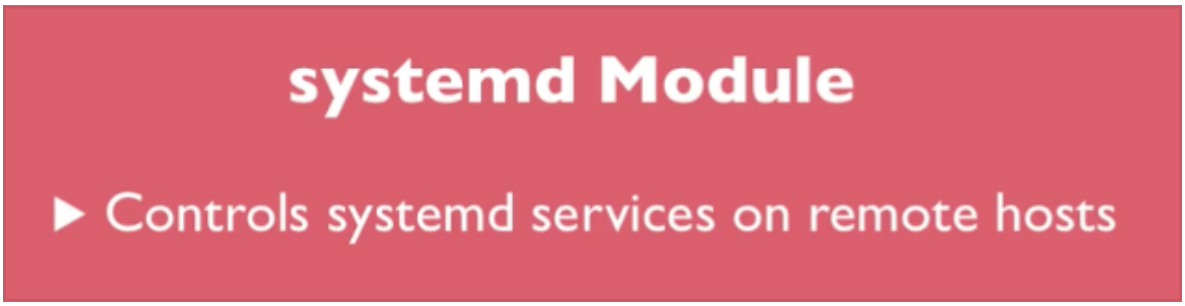
We have line 2, 7, 8. But docker is not running yet, so we nee to map line 3.

```
[ec2-user@ip-10-0-10-56 ~]$ docker pull redis
Using default tag: latest
Cannot connect to the Docker daemon at unix:///var/run/docker.sock. Is the docker daemon running?
[ec2-user@ip-10-0-10-56 ~]$
```

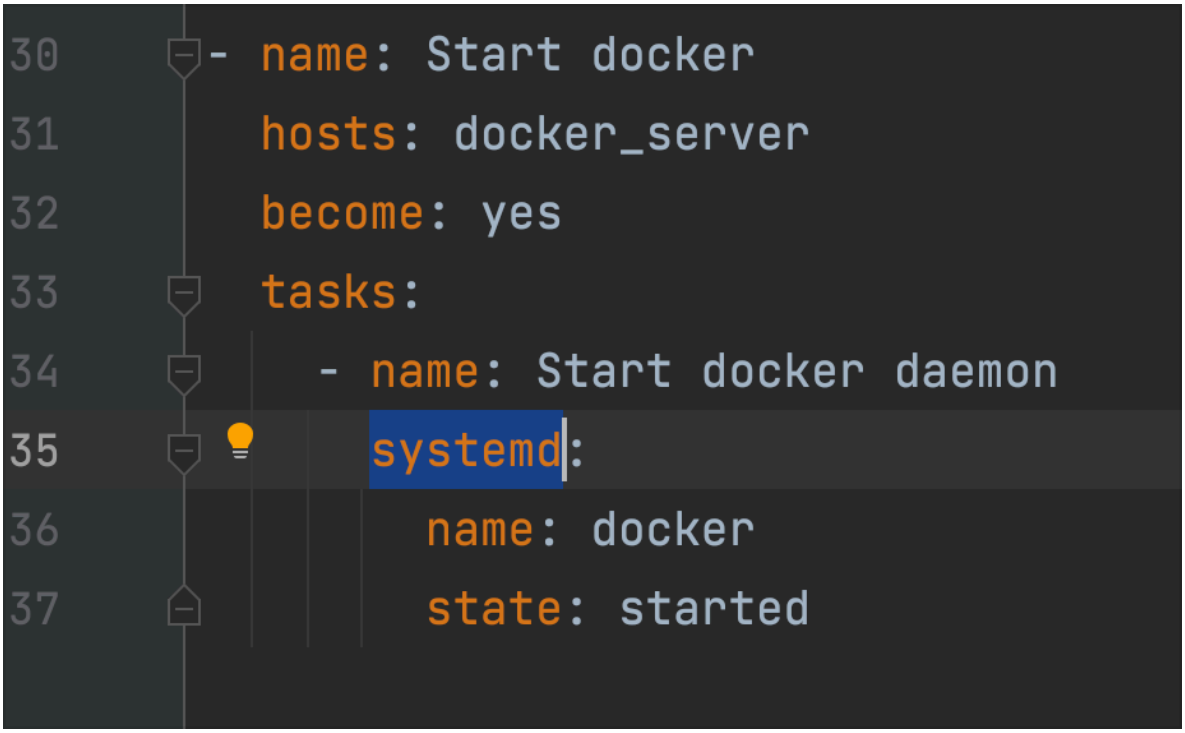
Write 3rd play

Start docker daemon

And we will need root user to run docker and the module ‘systems’ module to map ‘systemctl’



And the play looks like this



Execute the playbook
-> ansible-playbook deploy-docker.yaml

```
TASK [Gathering Facts] *********************************************************************************************
ok: [35.179.115.76]

TASK [Start docker daemon] *****************************************************************************************
changed: [35.179.115.76]

PLAY RECAP *********************************************************************************************************
35.179.115.76              : ok=8    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

And I check in the Ec2 instance

```
[ec2-user@ip-10-0-10-56 ~]$ docker pull redis
Using default tag: latest
Cannot connect to the Docker daemon at unix:///var/run/docker.sock. Is the docker daemon running?
[ec2-user@ip-10-0-10-56 ~]$ docker pull redis
Using default tag: latest
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Post "http://%2Fvar%2Frun%2Fdocker.sock/v
1.44/images/create?fromImage=redis&tag=latest": dial unix /var/run/docker.sock: connect: permission denied
[ec2-user@ip-10-0-10-56 ~]$
```

We can see docker is running but we have an error, a different error, we are having now permission issues. We should use sudo

-> sudo docker pull redis

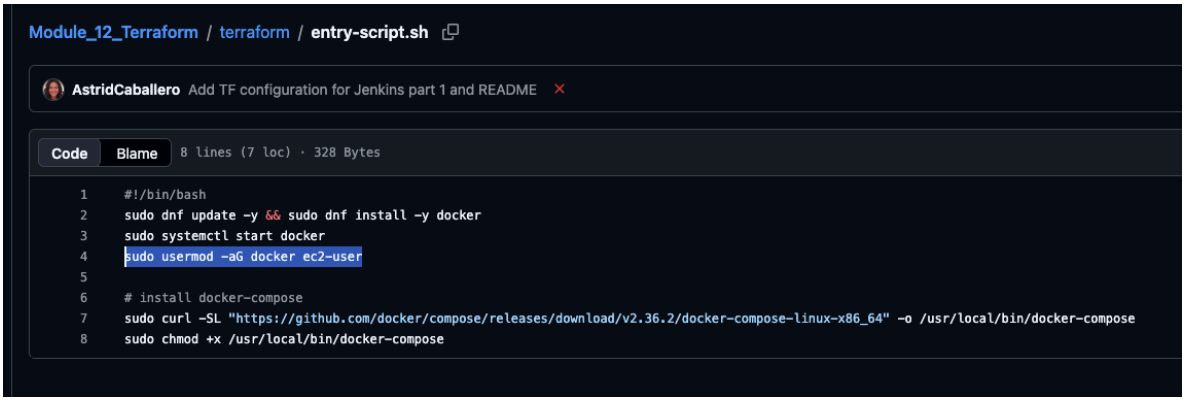
```
[ec2-user@ip-10-0-10-186 ~]$ sudo docker pull redis
Using default tag: latest
latest: Pulling from library/redis
e1caac4eb9d2: Extracting [===>                                              ]  2.064MB/29.12MB
7469c6c5b625: Download complete
a3d1b68c4a62: Download complete
152cbe749752: Download complete
7218480dfba1: Downloading [=====>                                           ]  2.149MB/20.83MB
e61c48a0d344: Waiting
4f4fb700ef54: Waiting
82adb0efabd8: Waiting
```

It will work using ‘sudo’ but not for the ec2-user because the user executing a docker command needs to be part of ‘dock’ group and ec2-user is not part of that group yet.

Write 4th play

Add ec2-user to docker group

So we need to map line 4 to ansible



To add ec2-user to the group ‘docker’
-aG -> add to group

Once we add ec2-user to the group ‘docker’ we can execute docker commands without ’sudo’ keyword

The new play needs to use ‘root’ user to add another user to a group

```
39  - name: Add ec2-user to docker group
40    hosts: docker_server
41    become: yes
42    tasks:
43      - name: Add ec2-user to docker group
44        user:
```

And we will use the ‘user’ module. With this module we can create a new user and also add a user to a group or groups.

Currently ec2-user belong to 4 groups:

```
[ec2-user@ip-10-0-10-56 ~]$ groups
ec2-user adm wheel systemd-journal
[ec2-user@ip-10-0-10-56 ~]$
```

So we want to append ‘docker’ group to the list of groups that the user belongs to:

```
39  - name: Add ec2-user to docker group
40    hosts: docker_server
41    become: yes
42    tasks:
43      - name: Add ec2-user to docker group
44        user:
45          name: ec2-user
46          groups: docker
47          append: yes
```

Manually we test the command
-> sudo usermod -aG docker ec2-user

```
[ec2-user@ip-10-0-10-186 ~]$ groups
ec2-user adm wheel systemd-journal
[ec2-user@ip-10-0-10-186 ~]$ sudo usermod -aG docker ec2-user
[ec2-user@ip-10-0-10-186 ~]$ groups
ec2-user adm wheel systemd-journal
[ec2-user@ip-10-0-10-186 ~]$
```

Note:
And after we add the user to the group we check the list and we can't see it.

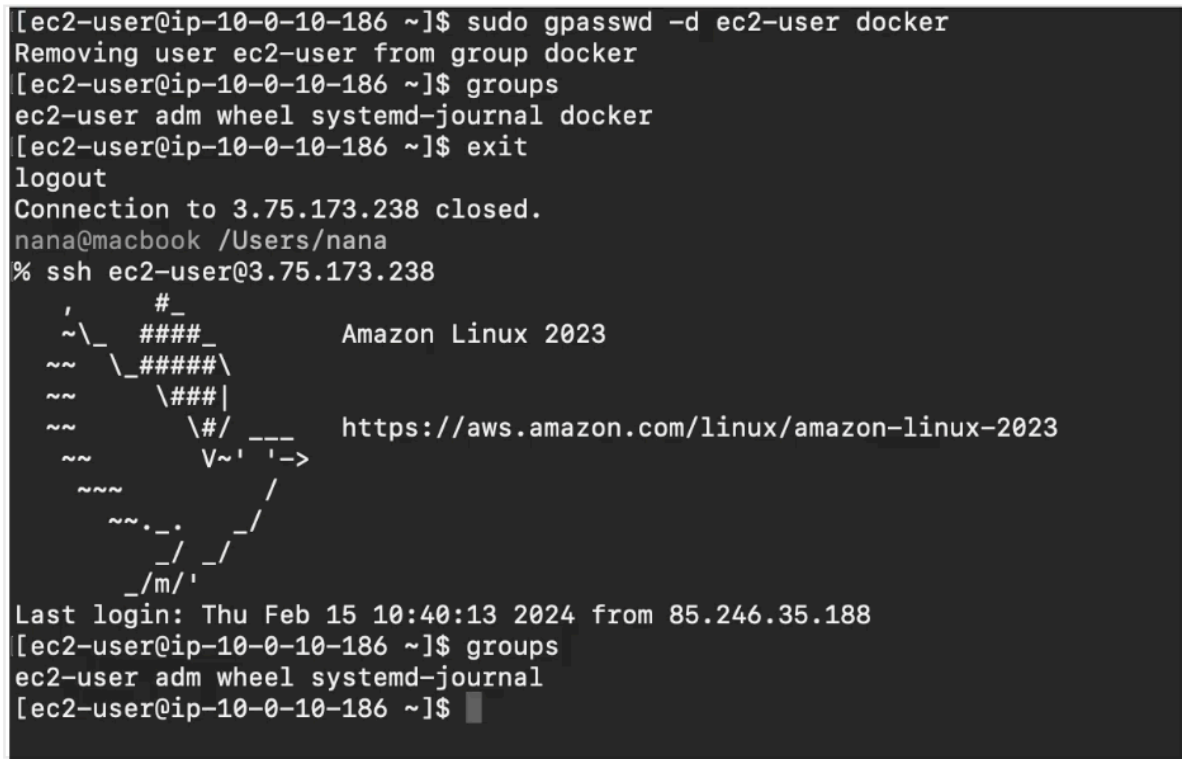
We need to log out and reconnect to be able to see it:

```
[ec2-user@ip-10-0-10-186 ~]$ groups
ec2-user adm wheel systemd-journal
[ec2-user@ip-10-0-10-186 ~]$ exit
logout
Connection to 3.75.173.238 closed.
nana@macbook /Users/nana
% ssh ec2-user@3.75.173.238
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Feb 15 10:33:56 2024 from 85.246.35.188
[ec2-user@ip-10-0-10-186 ~]$ groups
ec2-user adm wheel systemd-journal docker
[ec2-user@ip-10-0-10-186 ~]$
```

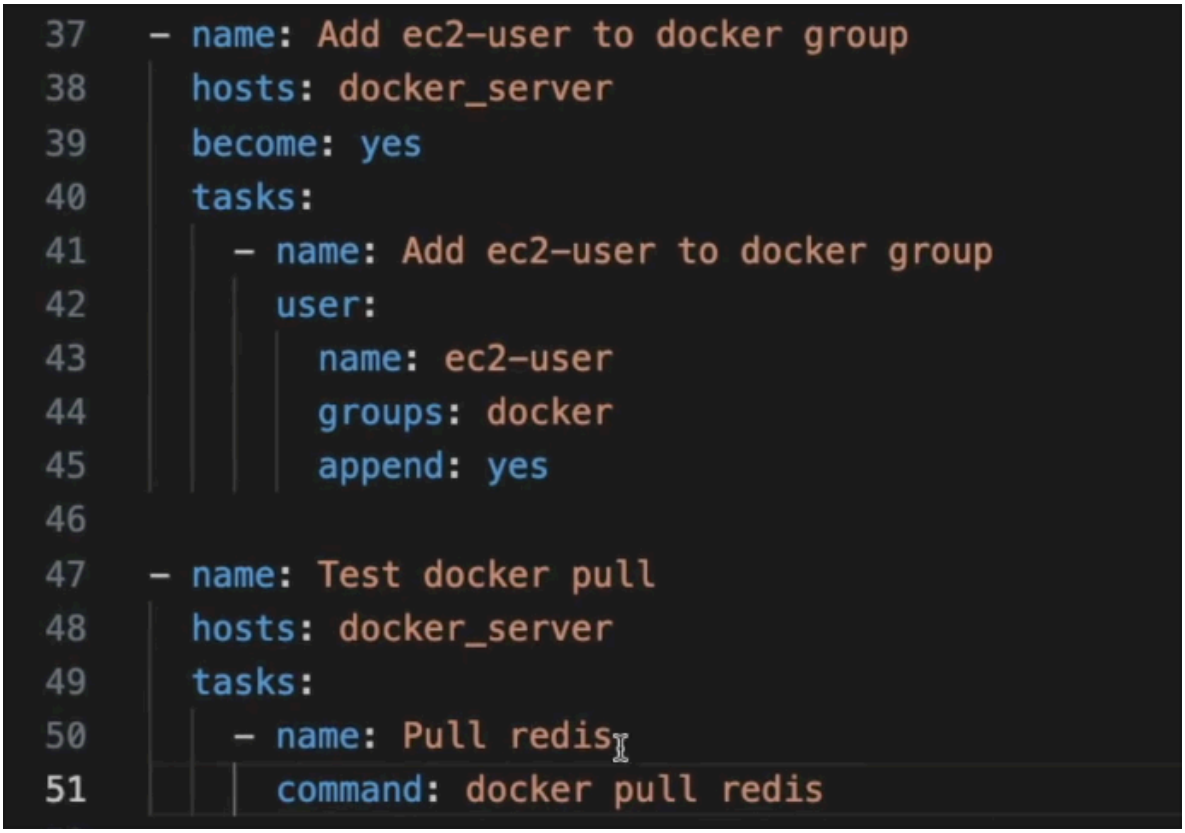
Now we can see it.

And we will have the same problem with our playbook, because we are doing all the tasks in the same section.

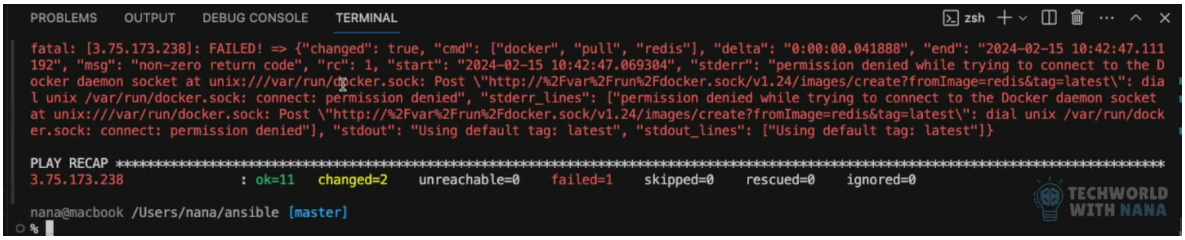
So Nana removes ec2-user from the 'docker' group



And creates a new play to test and show that we will have the problem



And she gets the permission error:



Ansible has a module called ‘meta’ to exit the session and reconnect to a new one:

```yaml
- name: Add ec2-user to docker group
  hosts: docker_server
  become: yes
  tasks:
    - name: Add ec2-user to docker group
      user:
        name: ec2-user
        groups: docker
        append: yes
    - name: Reconnect to server session
      meta: reset_connection

- name: test docker pull
  hosts: docker_server
  tasks:
    - name: Pull redis
      command: docker pull redis
```

Now execute the playbook:
-> ansible-playbook deploy-docker.yaml

```
TASK [Gathering Facts] *********************************************************************************************
ok: [35.179.115.76]

TASK [Add ec2-user to docker group] ********************************************************************************
changed: [35.179.115.76]

TASK [Reconnect to server session] *********************************************************************************

PLAY [test docker pull] ********************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [35.179.115.76]

TASK [Pull redis] **************************************************************************************************
changed: [35.179.115.76]

PLAY RECAP *********************************************************************************************************
35.179.115.76              : ok=12   changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

No errors this time ✅

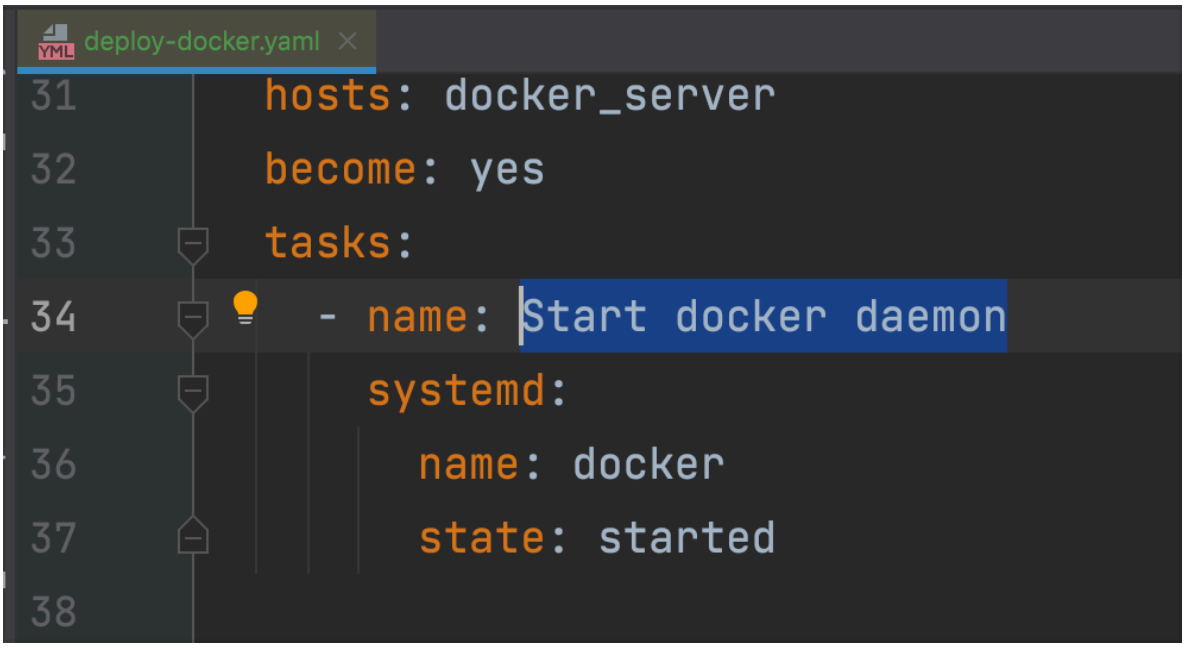
And check the server
-> docker images | grep redis

```
[ec2-user@ip-10-0-10-56 ~]$ docker images | grep redis
redis        latest    0458cdd27215   2 days ago   146MB
[ec2-user@ip-10-0-10-56 ~]$
```

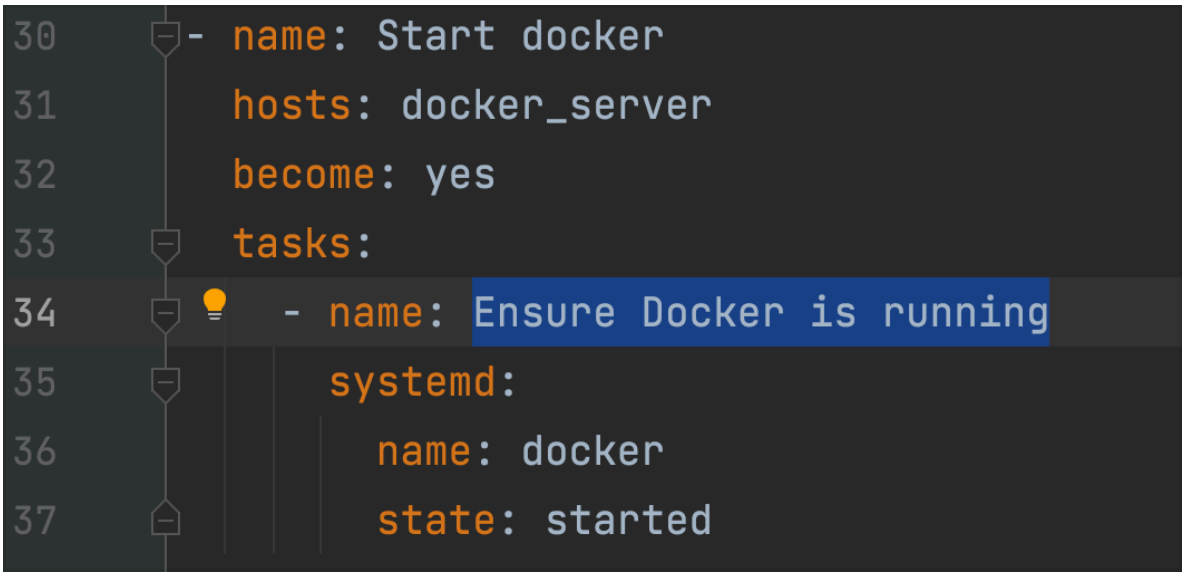
Refactor the playbook

In Ansible the standard is to name the tasks with a sentence that reflects the outcome of the task.

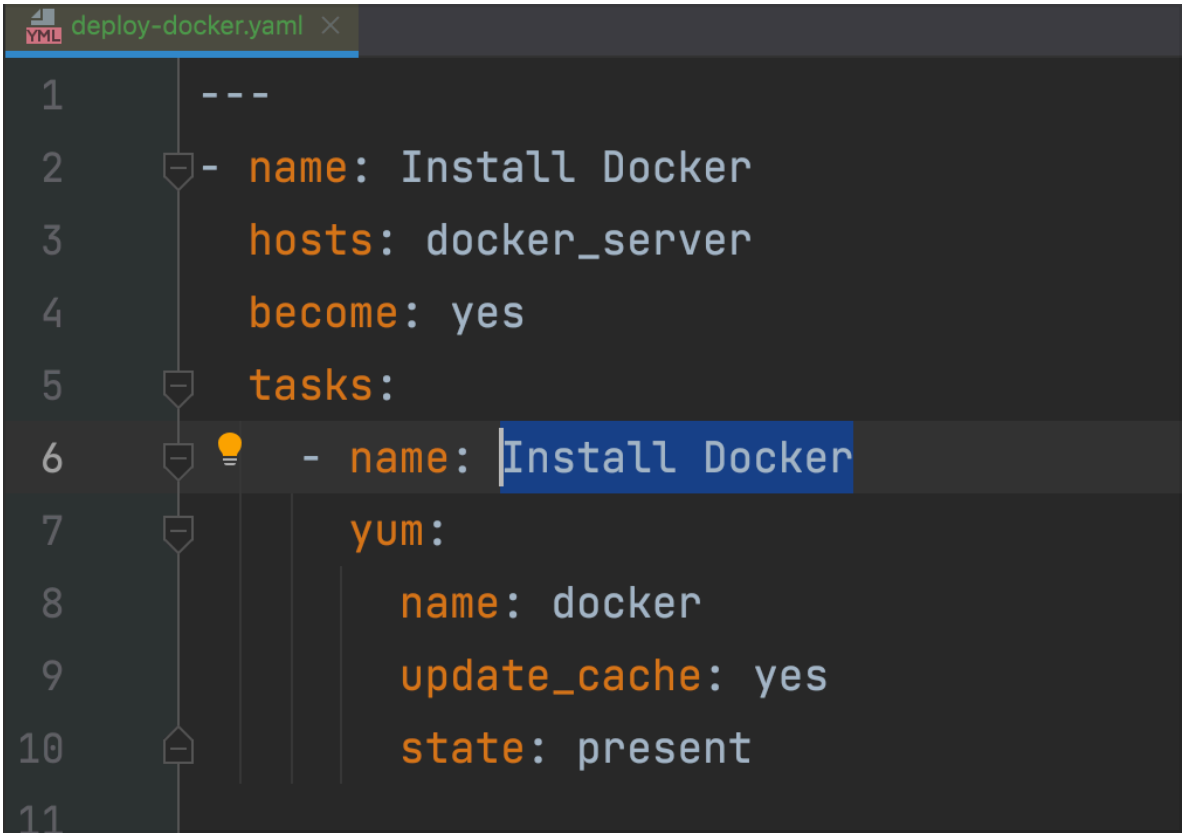
So instead of ‘Start docker daemon’



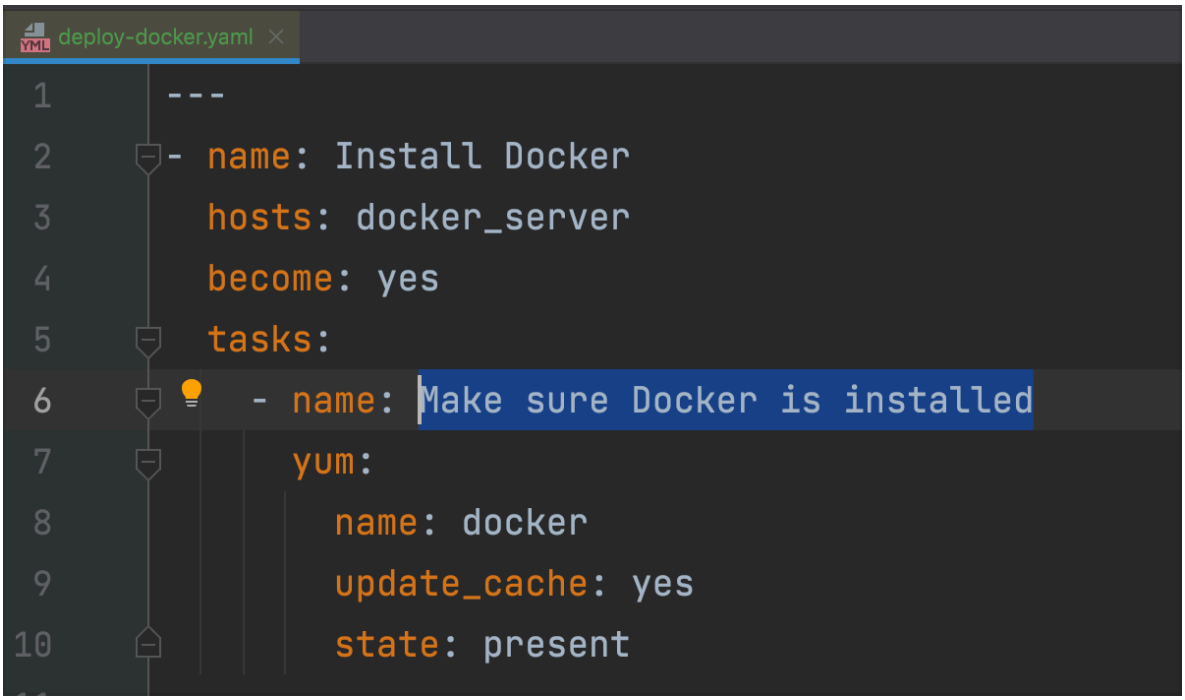
Is better ‘Ensure Docker is running’ as it will match the state: started



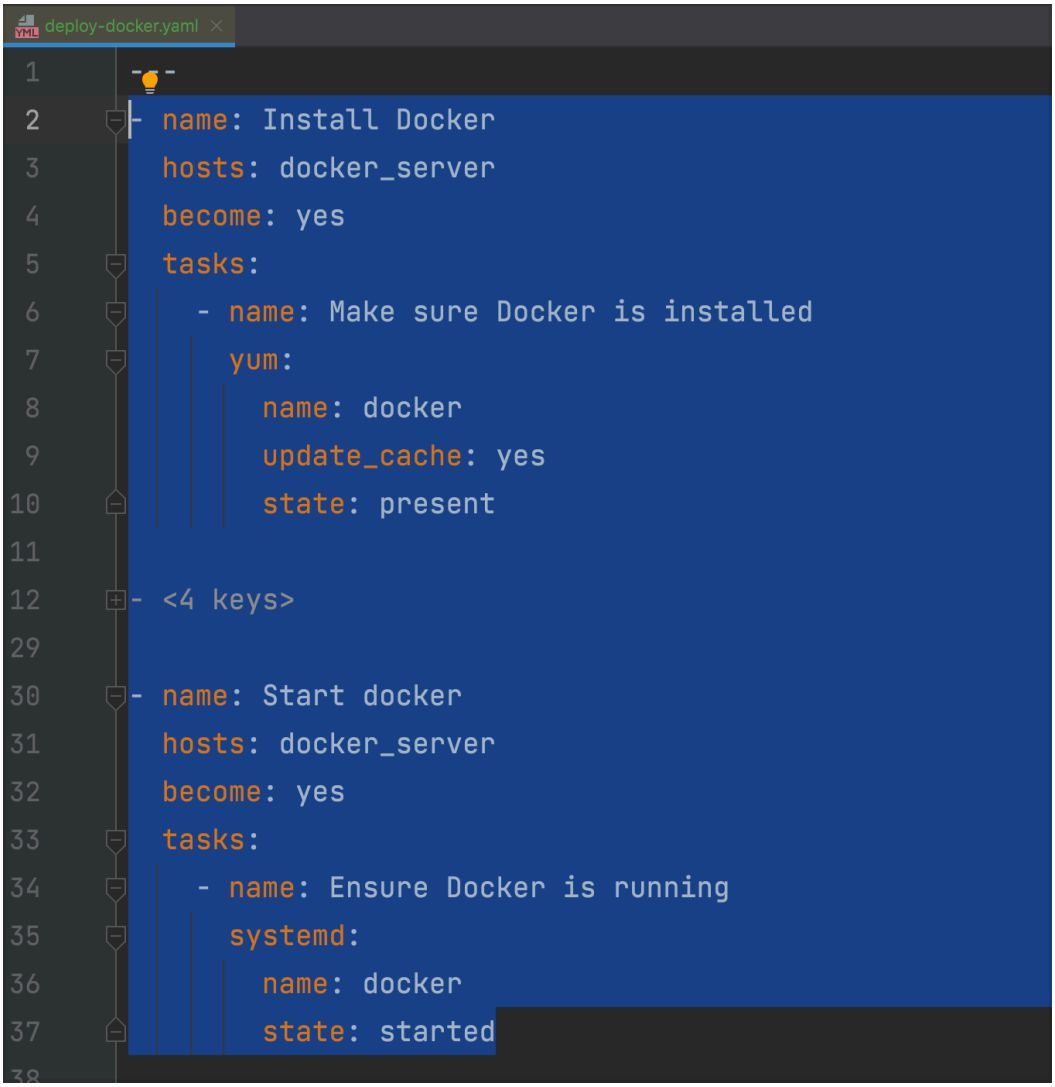
Same for ‘Install Docker’



Is better ‘Make sure Docker is installed’



We can also can start the docker deamon after we installed docker and it also make sense to have them in the same play as both need root user, so the two below



Become one play

deploy-docker.yaml

```yaml
---
- name: Install Docker
  hosts: docker_server
  become: yes
  tasks:
    - name: Make sure Docker is installed
      yum:
        name: docker
        update_cache: yes
        state: present
    - name: Ensure Docker is running
      systemd:
        name: docker
        state: started

- <4 keys>
#
#- name: Start docker
#  hosts: docker_server
#  become: yes
#  tasks:
```